



Enhancing Multimodal Emotion Recognition via Multi-feature Encoding and Attention-Based Fusion

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Abstract. Multimodal emotion recognition has attracted growing interest due to its importance in human-computer interaction, remote education, and healthcare. This paper proposes a novel multimodal emotion recognition framework that integrates rich audio and visual feature extraction with an attention-based fusion strategy. For audio, we extract three complementary feature types: semantic embeddings from Wav2Vec2, MFCC features, and statistical acoustic descriptors such as pitch, energy, and rhythm. These are aligned and fused via a BiLSTM to capture temporal dependencies. For video, we propose a ResNet50-BiLSTM architecture that combines deep residual learning and sequential modeling to extract expressive spatiotemporal features from facial sequences. To enhance multimodal synergy, we introduce a feature-level fusion mechanism based on multi-head attention, allowing the model to adaptively weigh contributions across modalities. Experiments conducted on the MELD and IEMOCAP datasets demonstrate that our model significantly outperforms baselines in both accuracy and robustness. Furthermore, ablation studies show that the attention-based fusion strategy significantly improves performance in unbalanced data settings. Our findings suggest that the proposed framework effectively captures diverse emotional cues from speech and visual expressions, and offers a practical and generalizable approach for real-world multimodal emotion recognition tasks.

Keywords: Emotion Recognition · Feature Extraction · Multimodal Fusion · Attention Mechanism

1 Introduction

Emotion plays a vital role in human communication, influencing decision-making, social interaction, and cognition [1]. As human-computer interaction systems become increasingly prevalent, enabling machines to accurately recognize and respond to human emotions is a key goal in affective computing [2].

Conventional unimodal approaches often relied on single-modal data such as speech, facial expressions, or physiological signals. However, these approaches are susceptible to noise contamination and information scarcity, along with lacking contextual relevance, resulting in reduced robustness and biased emotional interpretation.

In fact, human emotion perception depends on multiple modalities [3], integrating auditory, visual, and contextual cues. For instance, if a stressed friend replies “I’m doing great” while looking frustrated, a unimodal speech-based system may incorrectly predict a positive emotion. Only by integrating visual and contextual signals can the true emotional state be recognized [4].

Multimodal emotion recognition has advanced significantly. For example, Zhao et al. proposed VAANet [5], integrating spatial, channel, and temporal attention mechanisms in a CNN. Avro et al. introduced EmoTech [6], combining low-level audio and text features with a hybrid BiLSTM-CNN. However, challenges such as feature redundancy and insufficient cross-modal fusion remain [7].

To address these, we propose a multimodal framework inspired by human perception. We design unimodal encoders: an audio module combining three types of acoustic features processed via BiLSTM, and a video module using ResNet-BiLSTM to capture spatiotemporal facial dynamics. A cross-modal attention mechanism then fuses the extracted features.

Experiments on MELD and IEMOCAP validate the effectiveness of our framework, which outperforms state-of-the-art methods in overall accuracy and minority-class performance.

2 Methods

Our multimodal emotion recognition framework comprises three key components: (1) a BiLSTM-based audio module that integrates and models temporal dependencies across three complementary audio features; (2) a ResNet50-BiLSTM video module for extracting deep spatiotemporal features from facial sequences; and (3) a multi-head attention-based fusion module that adaptively combines modalities at the feature level.

2.1 BiLSTM-Based Multi-feature Audio Emotion Recognition

The structure of the proposed BiLSTM-based audio emotion recognition module is illustrated in Fig. 1. We first extract **Mel-frequency cepstral coefficients (MFCCs)** using the TorchAudio library to capture low-level spectral features. Next, we employ the pre-trained Wav2Vec2 model to obtain **semantic-level audio embeddings**. In parallel, we use the Librosa library to compute various prosodic and statistical descriptors, including **spectral centroid**, **zero-crossing rate**, **pitch**, **root mean square (RMS) energy**, and **rhythm**. Finally, these features are fed into a BiLSTM network for temporal sequence

modeling and contextual information extraction across feature dimensions. We note that the choice of libraries is intentional: Torchaudio is used for its seamless integration with the PyTorch ecosystem and GPU acceleration during MFCC extraction, which significantly improves preprocessing efficiency. Librosa, on the other hand, is employed for its comprehensive suite of hand-crafted prosodic and statistical descriptors beyond MFCCs. This combination leverages the respective strengths of each library.

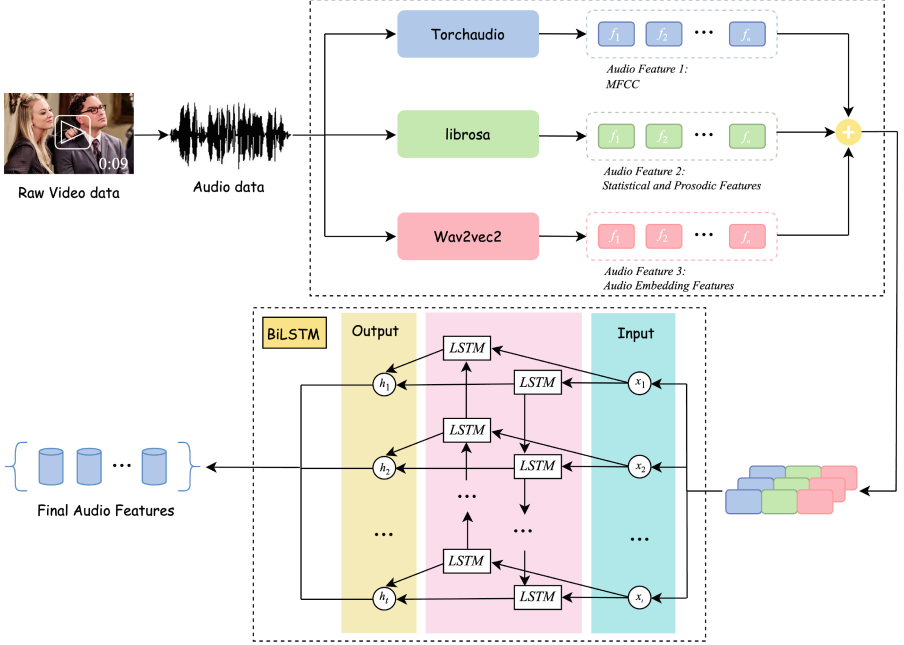


Fig. 1. Architecture of the audio feature extraction model

Multi-feature Audio Feature Extraction Model. Wav2Vec2 is a self-supervised model based on the Transformer architecture, consisting of a Feature Encoder and a Transformer Context module. The former maps each 20ms audio segment into a low-level feature vector, while the latter captures long-term dependencies and outputs semantic audio embeddings. The MFCC features approximate the human cochlea’s nonlinear frequency response and describe the spectral envelope of speech signals. The statistical audio features extracted via Librosa provide explicit acoustic cues related to emotional expression.

These three types of features are complementary to each other, enabling deep exploration of the intrinsic structure of audio signals, and improving both the robustness and expressive power of the system.

BiLSTM-Based Audio Feature Fusion Model. Let the three types of extracted features be:

- $\mathbf{V}_1$: Wav2Vec2-based semantic embedding (dimension 1×512)
- $\mathbf{V}_3$: MFCC features (dimension 1×25)
- $\mathbf{V}_4$: Statistical audio features (dimension 1×25)

We first project $\mathbf{V}_3$ and $\mathbf{V}_4$ to the same dimension as $\mathbf{V}_1$ via fully connected layers:

$$\mathbf{V}'_3 = \text{FC}_1(\mathbf{V}_3), \quad \mathbf{V}'_4 = \text{FC}_2(\mathbf{V}_4) \quad (1)$$

where $\mathbf{V}'_3 \in \mathbb{R}^{1 \times 512}, \mathbf{V}'_4 \in \mathbb{R}^{1 \times 512}$. The three features are fused by weighted summation:

$$\mathbf{F}_{\text{audio}} = \alpha \mathbf{V}_1 + \beta \mathbf{V}'_3 + \gamma \mathbf{V}'_4 \quad (2)$$

where $\alpha + \beta + \gamma = 1$. Then, the vector is reshaped to match the expected BiLSTM input format:

$$\mathbf{F}_{\text{audio}}^{(t)} = \text{Unsqueeze}(\mathbf{F}_{\text{audio}}) \quad (3)$$

where $\mathbf{F}_{\text{audio}}^{(t)} \in \mathbb{R}^{1 \times 1 \times 512}$. It is then passed through a bidirectional LSTM layer:

$$\vec{\mathbf{h}} = \text{LSTM}_{\text{fwd}}(\mathbf{F}_{\text{audio}}^{(t)}), \quad \overleftarrow{\mathbf{h}} = \text{LSTM}_{\text{bwd}}(\mathbf{F}_{\text{audio}}^{(t)}) \quad (4)$$

The final audio representation is obtained by concatenating forward and backward outputs:

$$\mathbf{z}_{\text{audio}} = [\vec{\mathbf{h}}; \overleftarrow{\mathbf{h}}] \quad (5)$$

In this equation, $\mathbf{z}_{\text{audio}} \in \mathbb{R}^{1 \times 512}$

2.2 BiLSTM-Based Video Emotion Recognition Module

The architecture of the BiLSTM-based video emotion recognition module is illustrated in Fig. 2.

It consists primarily of a ResNet-50 backbone and a BiLSTM layer for temporal modeling. We first apply an entropy-based frame selection algorithm to extract a fixed number of representative frames from raw video data. These frames are selected either at uniform time intervals or based on content-aware criteria, ensuring that frames with the highest information content are retained. The selected frames are then preprocessed and passed through a pre-trained ResNet-50 network, which extracts deep spatial features from each frame and outputs corresponding frame-level feature vectors. These vectors are fed sequentially into a BiLSTM network to model temporal dependencies and refine the emotional context. By processing the sequence in both forward and backward directions, the BiLSTM captures bidirectional relationships and generates the final video-level feature representation.

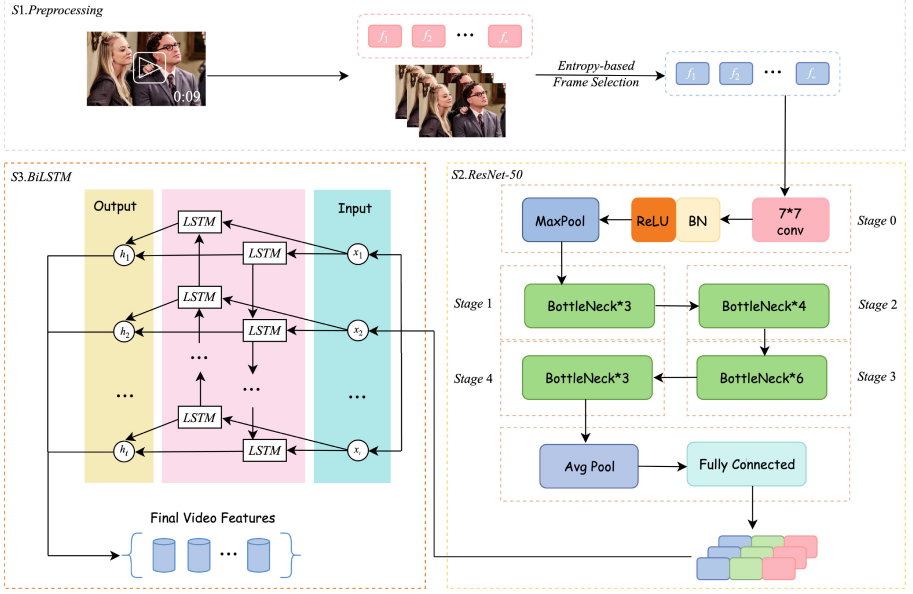


Fig. 2. Architecture of the video feature extraction model

Entropy-Based Frame Selection. To reduce redundancy and focus on highly informative content, we applied an entropy-based frame selection algorithm to each video segment. Specifically, a fixed number of representative frames were extracted either at fixed time intervals or based on content-aware entropy scoring. This strategy ensures that the most informative frames are retained, thereby improving model efficiency and reducing noise interference.

ResNet-50-Based Feature Extraction Model. The selected high-information-content frames are preprocessed and fed into a ResNet-50 network for high-level feature extraction. ResNet-50 employs residual units with skip connections, where the input is added directly to the output. This residual learning pathway facilitates gradient flow during backpropagation, mitigates vanishing gradients, and enables the training of deeper networks without accuracy degradation. The formula for the residual module is:

$$y = H(x, \mathbf{W}_H) + x \quad (6)$$

Specifically, we first apply an initial convolutional layer to extract low-level features from the input image, expressed as:

$$\mathbf{Y} = \text{Conv2D}(\mathbf{X}, \mathbf{W}_1) + \mathbf{b}_1 \quad (7)$$

where $\mathbf{Y}$ is the output feature map, $\mathbf{X}$ is the input image, $\mathbf{W}_1$ represents the 7×7 convolutional kernel weights with stride 2 and 64 output channels, and $\mathbf{b}_1$ denotes the bias term.

Next, the output feature map is passed through a max pooling layer to reduce its spatial dimensions by selecting the maximum value in each local region:

$$\mathbf{Z} = \text{MaxPool2D}(\mathbf{Y}) \quad (8)$$

where $\mathbf{Z}$ is the output feature map and $\mathbf{Y}$ is the input feature map.

ResNet-50 is composed of four stages, containing 3, 4, 6, and 3 residual blocks respectively. Each block employs skip connections that span two to three convolutional layers, establishing direct gradient flow pathways and improving the efficiency of parameter updates. This design helps mitigate the vanishing gradient problem in deep networks. The residual block is computed as:

$$F(\mathbf{X}) = \text{Conv2D}(\text{ReLU}(\text{Conv2D}(\text{ReLU}(\text{Conv2D}(\mathbf{X}, \mathbf{W}_2)), \mathbf{W}_3)), \mathbf{W}_4) \quad (9)$$

where $\mathbf{W}_2$, $\mathbf{W}_3$, and $\mathbf{W}_4$ correspond to 1×1 , 3×3 , and 1×1 convolutional kernels, respectively.

The skip connection is implemented as:

$$\mathbf{Y} = F(\mathbf{X}) + \text{Conv2D}(\mathbf{X}, \mathbf{W}_s) \quad (10)$$

where $\mathbf{W}_s$ is a 1×1 convolutional kernel used to align the input dimensions.

After passing through multiple stacked residual blocks, the resulting feature map $\mathbf{Z}'$ undergoes global average pooling over its spatial dimensions. This operation aggregates global information without introducing additional parameters and helps prevent overfitting. The formula is:

$$\mathbf{Z}_{\text{pool}} = \frac{1}{H' \times W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} \mathbf{Z}'_{i,j} \quad (11)$$

where $\mathbf{Z}_{\text{pool}}$ is the global feature vector with a shape of $1 \times 1 \times 2048$.

BiLSTM-Based Video Feature Fusion Model. Finally, the video features extracted by ResNet-50 are preprocessed and fed into a BiLSTM network, which hierarchically models both short- and long-range temporal patterns across frames across the frame sequence. By modeling the sequential structure, the BiLSTM extracts key time-dependent features and establishes contextual relationships within the video, yielding a temporally aggregated video embedding.

2.3 Multimodal Audio-Visual Emotion Recognition

We construct a multimodal emotion recognition model by combining the two unimodal feature extraction modules proposed in Sects. 2.1 and 2.2 with a feature-level fusion strategy based on a multi-head attention mechanism introduced in this section. The overall architecture is illustrated in Fig. 3.

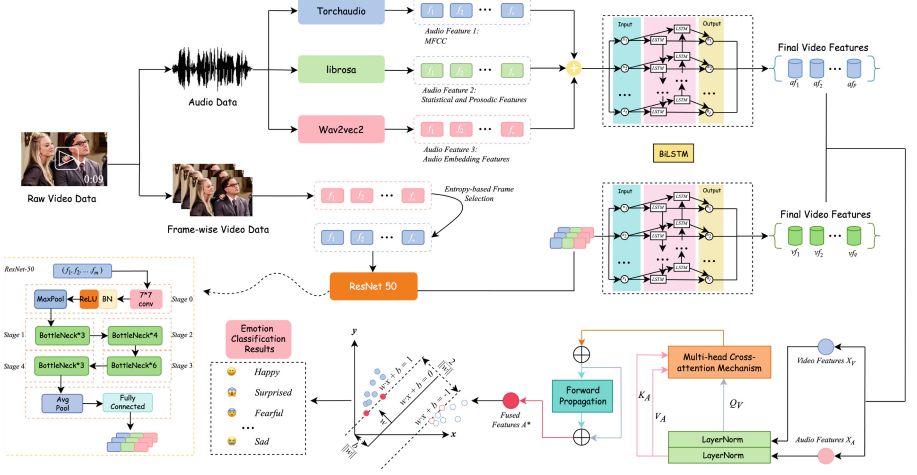


Fig. 3. Architecture of the multimodal audio-visual emotion recognition model

Feature-Level Fusion Based on Multi-head Attention Mechanism. We adopt a feature-level fusion strategy based on a multi-head attention mechanism to perform multimodal emotion recognition.

Specifically, given the audio feature vector $\mathbf{X}_A$ and video feature vector $\mathbf{X}_V$ obtained from the models described in Sects. 2.1 and 2.2, we first apply a LayerNorm layer to $\mathbf{X}_V$ to obtain the query vector $\mathbf{Q}_V$, and apply LayerNorm to $\mathbf{X}_A$ to obtain the key vector $\mathbf{K}_A$ and value vector $\mathbf{V}_A$. Then, $\mathbf{Q}_V$, $\mathbf{K}_A$, and $\mathbf{V}_A$ are passed through a multi-head cross-attention mechanism to compute the output vector $\mathbf{A}_X$. The attention mechanism is defined as follows:

$$\text{CrossAttention}(\mathbf{X}_A, \mathbf{X}_V) = \text{Softmax}\left(\frac{\mathbf{Q}_V \mathbf{K}_A^T}{\sqrt{d_V}}\right) \mathbf{V}_A \quad (12)$$

Here, $\mathbf{X}_V$ denotes the video feature vector, $\mathbf{Q}_V$ is the video query vector, and $\mathbf{X}_A$ is the audio feature vector. $\mathbf{K}_A$ and $\mathbf{V}_A$ represent the audio key and value vectors, respectively. We have $\mathbf{X}_A \in \mathbb{R}^{n \times d_A}$, $\mathbf{X}_V \in \mathbb{R}^{n \times d_V}$, where $\mathbf{W}^Q \in \mathbb{R}^{d_A \times d_k}$ and $\mathbf{W}^K \in \mathbb{R}^{d_V \times d_k}$ are learnable parameter matrices. Here, $\mathbb{R}$ denotes the set of real matrices, d_A is the dimension of the audio feature vector, d_V is the dimension of the value vector, and d_k is the dimension of the key vector.

The output attention vector $\mathbf{A}_X$ is added to the query vector $\mathbf{Q}_V$ and normalized to form an intermediate representation. This is passed through a feed-forward layer and added to the previous output to obtain the final fused feature vector $\mathbf{A}^*$.

This attention-based fusion strategy adaptively calibrates weights to different modalities, addressing feature redundancy and complementarity. It effectively captures long-range dependencies and contextual relevance while maintaining real-time performance. Additionally, it handles data imbalance and improves

the recognition accuracy for minority emotion classes, enhancing the robustness and precision of the overall system.

Emotion Classification with SVM. To effectively map the high-dimensional fused features to emotion categories and to ensure a fair comparison with several classic baseline methods that also utilize SVMs, we employ a Support Vector Machine (SVM) as the final classification module. SVMs, known for their strong performance in high-dimensional spaces and robustness to overfitting on smaller datasets, provide a powerful decision boundary for our extracted features.

In classification tasks, SVM maps the input data into a high-dimensional feature space and identifies a hyperplane that best separates the classes. Given a sample set $\mathbf{X} = \{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_N\}$ and corresponding labels $\mathbf{y} = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_N\}$, the decision hyperplane is defined by:

$$f(x) = \mathbf{w}^\top \mathbf{X} + b = 0 \quad (13)$$

where $\mathbf{w}$ is the weight vector perpendicular to the hyperplane, and b is the bias. The margin, defined as the distance between the closest samples on either side of the hyperplane (i.e., the support vectors), is:

$$d = \frac{2}{\|\mathbf{w}\|} \quad (14)$$

Support vectors satisfy $f(x) = 1$ for positive and $f(x) = -1$ for negative classes. The functional margin of a point $\mathbf{x}$ with label y is defined as:

$$\gamma = y(\mathbf{w}^\top \mathbf{x} + b) \quad (15)$$

This formulation allows SVM to generalize well even in high-dimensional spaces, making it suitable for distinguishing emotional categories based on fused feature representations.

3 Experiments

3.1 Evaluation of Single-Modal Emotion Recognition Models

Experimental Description. To evaluate the effectiveness of our designed single-modal audio and video emotion recognition models, we conducted experiments on the IEMOCAP and MELD datasets. Both datasets provide strictly synchronized multimodal data, enabling independent assessment of single-modal recognition models.

MELD (Multimodal EmotionLines Dataset) is an extension of the EmotionLines dataset for multimodal emotion recognition. It contains over 13,000 utterances from more than 1,400 dialogue segments of the TV show Friends. Each utterance is annotated with one of seven emotions: anger, disgust, sadness, joy, neutral, surprise, and fear, and includes corresponding audio, visual, and textual data. The distribution is shown in Table 1.

Table 1. Distribution Table of the MELD Dataset

Emotion Label	Training Set	Development Set	Test Set
anger	1109	153	345
disgust	271	22	68
fear	268	40	50
joy	1743	163	402
neutral	4710	470	1256
sadness	683	111	208
surprise	1205	150	281

IEMOCAP, released by the USC SAIL lab, includes scripted and improvised dyadic interactions by 10 professional actors. During data collection, synchronized audio, facial expressions, head movement, and hand gestures were recorded. The dataset totals approximately 12 h and provides fine-grained labels (including intensity and duration annotations), enabling multimodal temporal modeling research.

The experiments were conducted on a Linux system using Python 3.11, PyTorch 2.1.2, and CUDA 11.8.

We framed the emotion classification task as a seven-class problem (anger, disgust, sadness, joy, neutral, surprise, fear). The performance was evaluated using Accuracy, Precision, Recall, and F1-Score. Confusion matrices were used for visualization in selected experiments.

All models, including the unimodal encoders and the final multimodal network, are trained end-to-end to minimize the categorical cross-entropy loss between the predicted emotion probabilities and the ground-truth labels.

Evaluation of Audio Feature Extraction Model. We evaluated the proposed multi-feature audio extraction model (as described in Sect. 2.1) on the MELD dataset. The dataset was split into training and test sets in an 8:2 ratio. Figure 4. presents the confusion matrix results on the test set.

The results show good recognition accuracy for emotions such as anger, joy, and neutral, while performance on fear and sadness was relatively weaker. The overall accuracy reached 80.3%, indicating strong generalization capability.

To further validate the model’s effectiveness, we compared it against three commonly used audio feature extraction networks: RNN [8], 2D CNN [9], and 1D CNN-LSTM [10], under the same classifier. These established models provide a common baseline for ablative studies, offering a clear reference for measuring incremental gains. While newer architectures exist, our focus is to demonstrate improvements from our novel multi-feature and ResNet-BiLSTM design over these fundamental approaches. All baselines were implemented using original reported configurations (see cited works for details). The accuracy comparison is shown in Table 2.

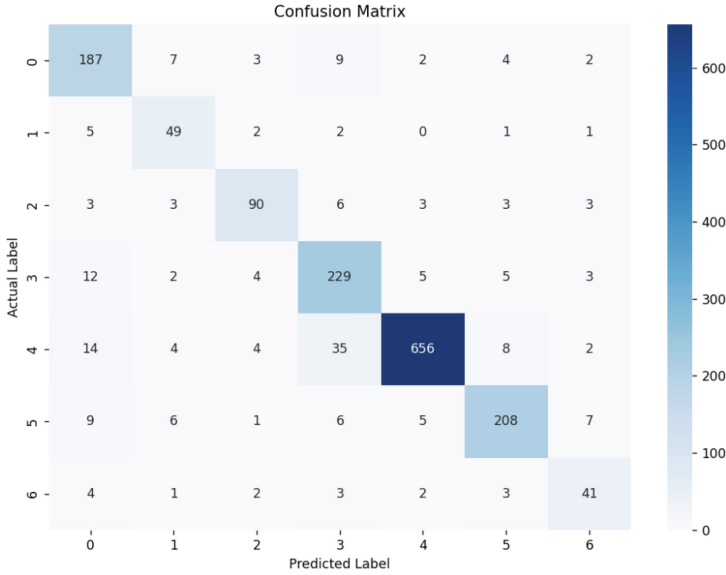


Fig. 4. Confusion Matrix of Audio-Based Emotion Recognition

Table 2. Comparison of Different Audio Feature Extraction Algorithms

Methods	Training Set	Test Set
RNN [8]	68.4%	68.3%
2D CNN [9]	69.2%	69.7%
1D CNN-LSTM [10]	73.9%	76.5%
Multi-audioFE	78.6%	80.3%

Our model achieved notable improvements: training accuracy increased by 10.2%, 9.6%, and 4.7% respectively, and test accuracy increased by 12%, 10.6%, and 3.8%, demonstrating clear advantages over baseline models in audio-based emotion recognition.

Evaluation of Video Feature Extraction Model. For the evaluation of video feature extraction, we first preprocessed the data to obtain 256×256 standard facial images. These frames were passed into the ResNet50-BiLSTM model proposed in Sect. 2.2 for feature extraction. We compared the performance with two classic video models: AlexNet [11] and GoogleNet [12]. The results are presented in Table 3.

Under the same emotion classifier, AlexNet achieved an average test accuracy of 68.1% on the MELD dataset, while GoogleNet slightly underperformed at 67.3%. Our ResNet50-BiLSTM model, benefiting from residual connections and deep temporal modeling, achieved 71.7% average accuracy.

Table 3. Comparison of Different Video Feature Extraction Algorithms

Methods	Accuracy	Precision	Recall	F1-Score
AlexNet [11]	68.1%	69.6%	66.7%	67.9%
GoogleNet [12]	67.3%	66.9%	66.9%	66.7%
ResNet50-BiLSTM	71.7%	72.3%	71.4%	71.8%

The results demonstrate that the proposed ResNet50-BiLSTM model is effective in capturing spatio-temporal features and yields superior performance in video-based emotion recognition.

3.2 Evaluation of Multimodal Emotion Recognition Model Performance

To evaluate the effectiveness of our proposed multimodal audio-visual emotion recognition model, we conducted experiments on the IEMOCAP and MELD datasets. The details of the datasets and experimental environment have already been described in Sect. 3.1 and are omitted here. We continued to use Accuracy, Recall, Precision, and F1-Score as evaluation metrics.

The MELD dataset was divided into training and test sets with an 8:2 ratio. The ROC curves for each emotion class on the test set are illustrated in Fig. 5..

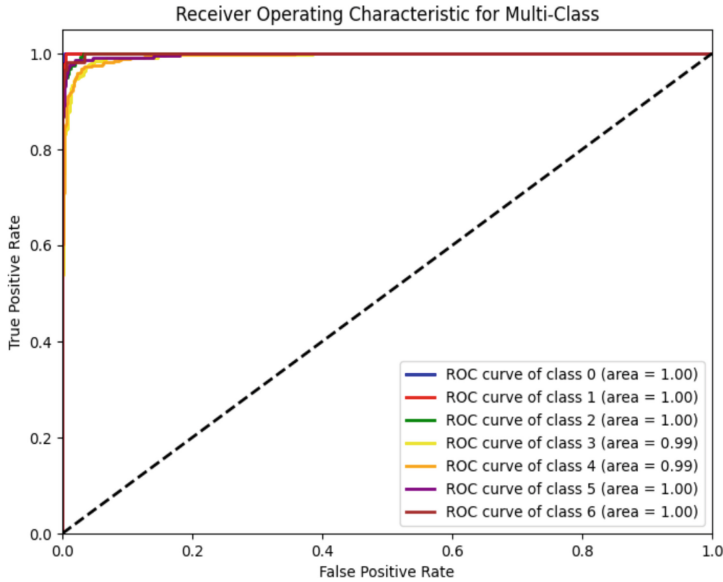


Fig. 5. ROC Curves of Multimodal Emotion Recognition

In the figure, each curve represents the ROC performance of an individual emotion category. The closer a curve is to the top-left corner, the better the model’s recognition ability for that emotion. As shown, most curves cluster near the top-left corner, indicating excellent classification performance.

The x-axis represents the false positive rate (FPR), which is the proportion of negative samples misclassified as positive. All emotion categories exhibit extremely low FPR, close to zero, indicating minimal false alarms. The y-axis denotes the true positive rate (TPR), with values close to 1, showing that the model correctly identifies nearly all positive samples.

Furthermore, the area under the curve (AUC) values for all seven emotion categories approach 1.00, further validating the strong performance of our model across all classes. These results preliminarily confirm the effectiveness of the proposed multimodal emotion recognition model.

Additionally, the confusion matrix of the model’s performance on the MELD test set is shown in Fig. 6.

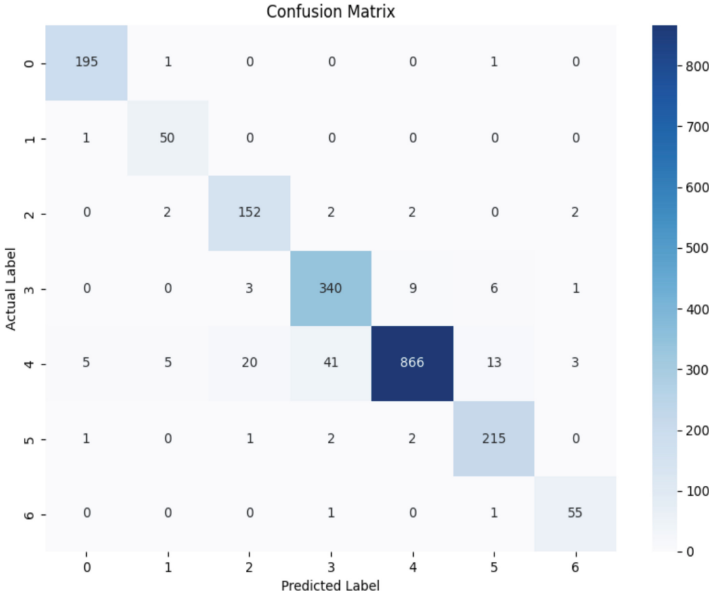


Fig. 6. Confusion Matrix of Multimodal Emotion Recognition

Based on the confusion matrix, we calculated the accuracy, precision, recall, and F1-Score for each emotion category, as presented in Table 4.

From the confusion matrix and evaluation metrics, we observe strong classification performance across all seven emotion categories, even with limited training samples. The average accuracy reaches 93.7%, and for the large-scale “neutral” class, accuracy remains above 90% despite a slight drop.

Table 4. Results Table of Multimodal Emotion Recognition

Emotion Label	Accuracy	Precision	Recall	F1-Score
anger	98.9%	96.5%	98.9%	97.7%
disgust	98.0%	89.2%	98.0%	93.4%
fear	95.0%	85.4%	95.0%	90.0%
joy	94.7%	86.0%	94.7%	90.2%
neutral	90.8%	97.6%	90.8%	94.1%
surprise	97.2%	96.8%	97.7%	97.2%
sadness	96.4%	88.7%	96.4%	92.4%
micro-average	93.7%	93.7%	93.7%	93.7%

These results demonstrate the high effectiveness of our model. By introducing an attention mechanism that dynamically allocates weights to different modalities, the model fully leverages complementary multimodal features, maintaining high classification accuracy.

Moreover, on the imbalanced test set, all emotion categories still achieve high accuracy and F1 scores, confirming the robustness and generalization ability of our model.

To further assess the generalizability of our approach, we conducted comparative experiments on the IEMOCAP dataset against several state-of-the-art multimodal emotion recognition models. The models and results are summarized in Table 5.

Table 5. Comparative Results of Multimodal Emotion Recognition Models

Model	Fusion Strategy	Accuracy	F1-Score
EF-LSTM	Feature-level Fusion	79.8%	79.1%
BERT	Feature-level Fusion	82.7%	83.6%
CNN-LSTM	Decision-level Fusion	82.9%	84.1%
Ours	Feature-level Fusion	79.8%	79.1%

As shown in Table 5, the proposed model, which integrates our audio extractor and ResNet50-BiLSTM visual extractor via multi-head attention-based fusion, outperforms EF-LSTM, BERT, and CNN-LSTM on the IEMOCAP dataset. Specifically, it achieves 10.5%, 7.6%, and 7.4% higher accuracy, and 9.4%, 4.9%, and 4.5% higher F1 scores, respectively.

These improvements stem from our modeling of intra-modal context, inter-modal interactions, and the incorporation of temporal dependencies to simulate the continuity of emotional expression.

To validate the effectiveness of our fusion strategy, Table 6 presents the results of ablation studies conducted on the MELD dataset.

Table 6. Ablation Study Results for Multimodal Emotion Recognition

Model	Fusion Strategy	Accuracy	F1-Score
Non-fused Feature Layer	Concatenated Vector	82.9%	84.1%
Fused Feature Layer	Attention-based Fusion	93.7%	93.5%

The results show that incorporating multi-head attention enables dynamic modality weighting and extraction of complementary information. Compared to models without fusion, our approach improves accuracy by 10.8% and F1-Score by 9.4%.

By integrating unimodal features using attention-based fusion, our model enhances inter-modal interactions and classification performance, validating the effectiveness of the proposed strategy.

4 Conclusions

In this paper, we proposed a multimodal emotion recognition framework that integrates deep feature extraction from both audio and video modalities with an attention-based fusion mechanism. The aim was to address the limitations of existing unimodal models in extracting high-quality features and to enhance the synergy between different modalities for improved classification performance.

Specifically, we designed a multi-feature audio extraction model that incorporates semantic embeddings from Wav2Vec2, MFCC features, and statistical acoustic descriptors. This design enables the model to capture audio cues from multiple perspectives, improving the robustness and expressiveness of emotional information. For visual features, we introduced a ResNet50-BiLSTM architecture that combines residual learning and temporal modeling to effectively extract rich emotional representations from facial video frames.

To bridge the gap between modalities, we implemented a multi-head attention-based feature-level fusion strategy. This fusion approach dynamically allocates attention weights across modalities, preserves the complementarity of multimodal data, and mitigates the risk of information loss during fusion. Experimental results on the MELD and IEMOCAP datasets demonstrate that our proposed framework achieves competitive performance across all major metrics and outperforms several state-of-the-art baselines.

Although the proposed method achieves promising experimental results, there remains room for further improvement. First, the current work primarily focuses on audio and visual modalities; future research could incorporate physiological signals such as EEG, skin conductance, and heart rate to better capture emotional states. For instance, Pardhan et al. combined DenseNet and LSTM to perform multimodal emotion recognition using physiological data [13]. Second, with the advancement of large multimodal models, pretrained networks offer a potential avenue to enhance feature extraction. Minoo Shayaninasab et

al. have explored multimodal emotion recognition using Transformer-based pre-trained models [14]. Finally, the Transformer architecture shows great promise in cross-modal modeling, enabling higher-level fusion strategies that improve generalizability and interpretability. For example, Zaidi et al. proposed a multimodal dual-attention Transformer for emotion recognition, capturing rich inter-modal emotional interactions [15].

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